

Esteban Fernando Cárdenas Andino

Padova, Italy

esteban.cardenas2001@gmail.com
GitHub Profile

Professional Summary

Data Physicist and Computational Scientist interested in complex physical systems and scalable data engineering. Currently pursuing a Master's in Physics of Data, with a proven track record in applying advanced statistical mechanics, machine learning, and high-performance computing. Strong background in interdisciplinary collaboration, technical leadership, and rapidly adapting to novel computational challenges in industry environments.

Education

Università degli Studi di Padova <i>Master's in Physics of Data</i> Specializing in computational physics, machine learning, stochastic processes, and statistical mechanics of complex systems.	October 2025 – Present <i>Padova, Italy</i>
Escuela Politecnica Nacional <i>Bachelor in Physics Cum Laude (9 semester program)</i>	June 2020 – December 2024 <i>Quito, Ecuador</i>

Technical Skills

Programming Languages: Python (Advanced), SQL, C++ (Intermediate), Fortran (Intermediate), R (Intermediate)
Machine Learning & Data Science: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, SciPy, U-Net, Gradient Boosting
Software Engineering & Tools: Docker, Docker Compose, Git, Jupyter Notebooks, Zsh, Relational Databases (MySQL)
Domain Expertise: High-Frequency Data Processing, Monte Carlo Simulations, Distributed Pipelines, Image/Phase Retrieval, Time-Series Analysis
Languages: Spanish (Native), English (Fluent)

Open Source & Data Engineering Projects

Comparative Machine Learning Benchmarking <i>Model Evaluation for Structured and Sequential Datasets</i> - Benchmarked Gradient Boosting Decision Trees against Neural Network architectures, evaluating performance metrics across tabular and time-series activity recognition datasets. - Orchestrated containerized environments using Docker Compose and executed complex MySQL queries with multi-table joins to manage and retrieve relational datasets efficiently.	Laboratory of Computational Physics <i>Python, PyTorch, Scikit-learn</i>
High-Frequency Tick Data Wrangler <i>Statistical Mechanics Approach to Market Microstructure</i> Engineered a high-performance pipeline processing millions of raw, high-frequency cryptocurrency trades (ticks) by applying advanced market microstructure theory to extract predictive, low-noise signals.	GitHub <i>Python, Jupyter</i>

EUV Phase Retrieval

GitHub

Digital Twin and U-Net Surrogate for Inverse Optical Phase Retrieval

Python, PyTorch

- Architected a PyTorch-based optical simulator featuring differentiable physics, Total Variation optimization, and Monte Carlo Uncertainty estimation.

Streaming Processing of QUAX Experiment Data

GitHub

Distributed Live Processing Pipeline

Jupyter Notebook

- Deployed a distributed live processing pipeline to efficiently handle high-throughput experimental data and enable rapid downstream analysis.

Phase-Transition Dynamics in 2D Lennard-Jones Fluid

GitHub

Non-equilibrium Thermodynamics Simulation

Python, PyTorch, CUDA

- Developed a GPU-accelerated simulation exploring phase transition dynamics, successfully demonstrating spontaneous symmetry breaking and macroscopic droplet formation.

Custom 8-Bit Breadboard Computer

Hardware Architecture

Logic Gate Level System Design

Electronics

- Constructed and debugged an 8-bit computer system based on Ben Eater's architecture, thoroughly analyzing circuit behavior, RAM module operations, and fetch cycle signals.
- Troubleshooted hardware-level memory volatility and optimized control word execution logic to guarantee stable processing performance.

Simulation of a Positron-induced Muon Source

GitHub

Monte Carlo Simulation

Jupyter Notebook

- Programmed a Monte Carlo simulation to accurately model the unpolarized scattering process $e^+e^- \rightarrow \mu^+\mu^-$ via a virtual photon mediator.

Professional & Research Experience

Research, Temporal Estimators of Peculiar GRBs

November 2023 – Present

Escuela Politecnica Nacional

Quito, Ecuador

- Engineered custom Python codebases to process and compute complex spectral and temporal parameters for large-scale astronomical datasets.
- Executed rigorous data analysis pipelines on transient phenomena using HEASoft and the Swift observatory database, translating raw signals into actionable quantitative metrics.

Data Analysis Intern (Astrophysics)

March 2024 – June 2024

Radboud University Postdoctoral Research / EPN

Quito, Ecuador

- Implemented advanced statistical modeling on gamma-ray bursts and transient phenomena, systematically identifying anomaly patterns in unstructured data.

Condensed Matter Laboratory Intern

October 2022 – March 2023

Escuela Politecnica Nacional

Quito, Ecuador

- Formulated and characterized graphene quantum dots utilizing precise hydrothermal synthesis methodologies.
- Quantified material property data using UV-VIS, FTIR, and Cyclic Voltammetry, ensuring high-fidelity outputs in physical measurements.

Publications & Presentations

- **Publication:** Noriega, N., Cárdenas, E. F., Chacón, J. A., & Vásquez, N. (2024). Temporal Estimators of Peculiar GRBs. *Journal of Physics: Conference Series* (Vol. 2796, Issue 1, p. 012001). <https://doi.org/10.1088/1742-6596/2796/1/012001>
- **Presentation:** Poster presentation at the 9th Colombian Meeting on High Energy Physics (9th ComHEP), *Unveiling the Enigmatic Nature of Peculiar Long GRBs in Swift's Third Catalog* (December 2024).

- **Presentation (Awarded):** Second place best poster at the International Congress of Physics (CIF) ESPOCH, Riobamba, *Unveiling the Enigmatic Nature of Peculiar Long GRBs* (October 2024).
- **Presentation:** Poster presentation at the XXXII IAU General Assembly, Cape Town, *Unveiling the Enigmatic Nature of Peculiar Long GRBs* (August 2024).
- **Presentation:** Poster presentation at the Physics meeting EPN, Quito, *Temporal estimators of peculiar GRBs* (October 2023).

Continuous Education & Leadership

- **Summer School in Statistics for Astronomers:** Center for Astrostatistics at Pennsylvania State University (June 2024) — Intensive training in Bayesian and non-Bayesian statistical methods.
- **Technical Leadership:** President of the Association of Physics Students at Escuela Politecnica Nacional (January 2024 – January 2025).
- **Event Organization:** Member of the organization committees for the Physics Meeting at EPN and the IV Workshop *Astronomia en los Andes*.
- **Awards:** Third place winner at the Dorothy Coding Challenge (ICTP); Second place at the IAPS PLANCKS national preliminaries.